ILLINOIS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Database Organization
Course CS-425 02

Homework No.2

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1.Given the Relation R (A,B,C,D,E) with functional dependencies F ={CE → D, D→B, C→A}, find the following:

a) **All candidate keys?**
b) **Normal forms the Relation, R satisfies?**
c) **If not in 3NF, show the process to take it to 3NF?**
d) **Compute the canonical cover of the given function dep, F.**

$$R = (A, B, C, D, E)$$

$$F = \begin{Bmatrix} CE \to \mathrm{D} \\ D \to \mathrm{B} \\ C \to \mathrm{A} \end{Bmatrix}$$

Finding all candidate keys:

To find all candidate keys, we need to determine the minimal sets of attributes that can uniquely determine all other attributes in the relation.

Using $C \to A$: $(C)^+ = \{C, A\}$ C determines A.	Using $D \to B$ $(D)^+ = \{D, B\}$ D determines B.	Using $CE \to D$ $(CE)^+ = \{C, E, D\}$ CE determines D.

Combining these to find potential candidate keys:

C alone cannot be a candidate key because it does not determine all attributes.

CE together can be a candidate key because $(CE)^+ = \{C, E, D, A, B\} = R$

a) Therefore, CE is a candidate key for the relation R.
b) Normal forms the Relation, R satisfies:

   To determine the normal forms the relation R satisfies:

- 1NF: The relation is in 1NF if all attributes are atomic. Assuming all attributes are atomic, the relation R satisfies 1NF.

- 2NF: The relation R is in 2NF if it is in 1NF and every non-prime attribute (attributes not part of any candidate key) is fully functionally dependent on the whole of every candidate key.
  Since R has a single candidate key CE, we need to check if $D \to B$ violates 2NF.
  $D \to B$ suggests a partial dependency (since D is not a superkey by itself), which violates 2NF.

c) 3NF: To be in 3NF, a relation must be in 2NF and no transitive dependencies should exist.
   The dependency $D \to B$ is a transitive dependency (through $CE \to D$ and $D \to B$).
   Taking the relation to 3NF: To normalize R into 3NF, we need to decompose it to eliminate the transitive dependency $D \to B$.
   $R1(C, E, D)$ with $CE \to D$
   $R2$(D,B) with D→B
   $R3$(C,A) with C→A

d) Find minimal cover: The canonical cover of F is the set of dependencies that is equivalent to F and minimal (no extraneous attributes or redundant dependencies).

New functional dependencies

$$F = \begin{Bmatrix} CE \rightarrow \text{D} \\ D \rightarrow \text{B} \\ C \rightarrow \text{A} \end{Bmatrix}$$

Step 1. Decomposition rule, no relationships to decompose on RHS.

Step 2: Eliminating extraneous attributes on LHS

$$CE \rightarrow \text{D}$$

Check C $(E)^+ = \{E\}$ C Is not extraneous	Check E $(C)^+ = \{C\}$ $= \{CA\}$ E Is not extraneous

No extraneous attributes are present.

Step 3. Eliminate redundant functional dependencies

First, eliminate any redundant functional dependencies and simplify each functional dependency to its minimal form.

Checking redundancy:

$CE \rightarrow \text{D}$: $(CE)^+ = \{CE\}$ $= \{CEA\}$ No redundancy found	$D \rightarrow \text{B}$: $(D)^+ = \{D\}$ No redundancy found	$C \rightarrow \text{A}$: $(C)^+ = \{C\}$ No redundancy found

Step 4: Apply Union. Combine dependencies and finalize the canonical cover. Now, combine the simplified dependencies: No combination in dependencies were necessary.

Canonical cover of F:

$$F_c = \begin{Bmatrix} CE \rightarrow \text{D} \\ D \rightarrow \text{B} \\ C \rightarrow \text{A} \end{Bmatrix}$$

2.Provided R(A, B,C,D,E,F,G), compute the canonical (minimal) cover of

F ={AD → BF, CD→ EGC, BD → F, E → D, F→C, D→F}

$$R = (A, B, C, D, E, F, G)$$

$$F = \begin{Bmatrix} AD \to BF \\ CD \to EGC \\ BD \to F \\ E \to D \\ F \to C \\ D \to F \end{Bmatrix}$$

Step 1. Decomposition rule

$$AD \to BF = \begin{matrix} AD \to B \\ AD \to F \end{matrix}$$

$$CD \to EGC = \begin{matrix} CD \to E \\ CD \to G \end{matrix} \; CD \to C$$

$$FDs = \begin{Bmatrix} AD \to B \\ AD \to F \\ CD \to E \\ CD \to G \\ CD \to C \\ BD \to F \\ E \to D \\ F \to C \\ D \to F \end{Bmatrix}$$

Step 2: Eliminating extraneous attributes on LHS

$$AD \to B$$

Check A $(D)^+ = \{D\}$ $= \{DF\}$ $= \{DFC\}$ $= \{DFCEG\}$ A Is not extraneous	Check D $(A)^+ = \{A\}$ D Is not extraneous

$$AD \to F$$

Check A $(D)^+ = \{D\}$ $= \{DF\}$ $= \{DFC\}$ $= \{DFCEG\}$ A Is not extraneous	Check D $(A)^+ = \{A\}$ D Is not extraneous

$CD \to E$ --Becomes $D \to E$

Check C $(D)^+ = \{D\}$ $= \{DF\}$ $= \{DFC\}$ C Is extraneous	Check D $(C)^+ = \{C\}$ D Is not extraneous

$CD \rightarrow G$ --Becomes $D \rightarrow G$

Check C $(D)^+ = \{D\}$ $= \{DF\}$ $= \{DFC\}$ C Is extraneous	Check D $(C)^+ = \{C\}$ D Is not extraneous

$CD \rightarrow C$ --Becomes $D \rightarrow C$

Check C $(D)^+ = \{D\}$ $= \{DF\}$ $= \{DFC\}$ C Is extraneous	Check D $(C)^+ = \{C\}$ D Is not extraneous

$BD \rightarrow F$

Check B $(D)^+ = \{D\}$ $= \{DF\}$ $= \{DFC\}$ $= \{DFCEG\}$ B Is not extraneous	Check D $(B)^+ = \{B\}$ D Is not extraneous

New set of FDs

$$FDs = \begin{Bmatrix} AD \rightarrow B \\ AD \rightarrow F \\ D \rightarrow E \\ D \rightarrow G \\ D \rightarrow C \\ BD \rightarrow F \\ E \rightarrow D \\ F \rightarrow C \\ D \rightarrow F \end{Bmatrix}$$

Step 3. Eliminate redundant functional dependencies

First, eliminate any redundant functional dependencies and simplify each functional dependency to its minimal form.

Checking redundancy:

AD → B: $(AD)^+ = \{AD\}$ $= \{ADF\}$ $= \{ADFC\}$ $= \{ADFCE\}$ No redundancy found	AD → F: $(AD)^+ = \{AD\}$ $= \{ADB\}$ $= \{ADBEGC\}$ $= \{ADBEGCF\}$ Since F was found AD → F is redundant	D → E: $(D)^+ = \{D\}$ $= \{DGCF\}$ No redundancy found
D → G: $(D)^+ = \{D\}$ $= \{DECF\}$ No redundancy found	D → C: $(D)^+ = \{D\}$ $= \{DEGF\}$ No redundancy found	BD → F: $(BD)^+ = \{BD\}$ $= \{BDCGEF\}$ Since F was BD → F is redundant
E → D: $(E)^+ = \{E\}$ No redundancy found	F → C: $(F)^+ = \{F\}$ No redundancy found	D → F: $(D)^+ = \{D\}$ $= \{DEG\}$ No redundancy found

Canonical Cover

$$Fc = \begin{Bmatrix} AD \to B \\ D \to E \\ D \to G \\ E \to D \\ F \to C \\ D \to F \end{Bmatrix}$$

Step 4: Apply Union rule to new set (minimal)

$$\begin{aligned} &D \to E \\ &D \to G = D \to EGF \\ &D \to F \end{aligned}$$

$$Fc = \begin{Bmatrix} AD \to B \\ D \to EGF \\ E \to D \\ F \to C \end{Bmatrix}$$

3. Suppose you are given a relation R=(A,B,C,D,E) with the following functional dependencies: {BC → ADE,D→B}

a) Find all candidate keys
b) Identify the best normal form the R satisfies

a) Finding all candidate keys:

A candidate key is a minimal set of attributes that can uniquely determine all other attributes in the relation.

Using $BC \to ADE$:

$BC+=\{B,C,A,D,E\}$ BC determines A,D,E.

Using $D \to B$:

$D+=\{D,B\}$

D determines B.

Now, let's combine these to find potential candidate keys:

BC can be a candidate key because $BC+= \{B, C, A, D, E\} =R$

Therefore, BC is a candidate key for the relation R.

b) Identify the best normal form the relation satisfies:

To determine the best normal form the relation R satisfies, we need to check for any violations of higher normal forms.

1NF (First Normal Form):

The relation R is in 1NF if all attributes are atomic. Assuming all attributes are atomic (single-valued), R satisfies 1NF.

2NF (Second Normal Form):

R is in 1NF. However, it does not satisfy 2NF because of the partial dependency D→B Here, B is part of the candidate key BC but is determined by D alone, which is not the entire candidate key. This makes B partially dependent on D, violating the requirement for 2NF where every non-prime attribute must be fully functionally dependent on every candidate key.

Third Normal Form (3NF):

For a relation to be in 3NF, it first needs to be in 2NF. Additionally, there should be no transitive dependencies involving non-prime attributes. R cannot be in 3NF because it does not satisfy 2NF due to the presence of a partial dependency.

Candidate keys: BC

Best normal form satisfied: R satisfies 1NF but does not satisfy 2NF or 3NF due to the transitive dependency

$D \rightarrow B$. Therefore, the best normal form that R satisfies is 1NF.

4. Given, R = {A,B,C,D,E,F,G,H} and F = {AC→G, D→EG, BC→D, CG→BD, ACD→B, CE→AG}.

Find the canonical cover of F.

$$R = (A, B, C, D, E, F, G, H)$$

$$FDs = \begin{Bmatrix} AC \rightarrow G \\ D \rightarrow EG \\ BC \rightarrow D \\ CG \rightarrow BD \\ ACD \rightarrow B \\ CE \rightarrow AG \end{Bmatrix}$$

Step 1. Decomposition rule

$$D \rightarrow EG = \begin{matrix} D \rightarrow E \\ D \rightarrow G \end{matrix}$$

$$CG \rightarrow BD = \begin{matrix} CG \rightarrow B \\ CG \rightarrow D \end{matrix}$$

$$CE \rightarrow AG = \begin{matrix} CE \rightarrow A \\ CE \rightarrow G \end{matrix}$$

Rewriting $ACD \rightarrow B$

Check AC $(D)^+ = \{D\}$ $= \{DEG\}$ AC Is not extraneous	Check D $(AC)^+ = \{AC\}$ $= \{ACG\}$ $= \{ACGBD\}$ Since D appeared, is extraneous and it becomes $AC \rightarrow B$

New FDs

$$FDs = \begin{Bmatrix} AC \rightarrow G \\ D \rightarrow E \\ D \rightarrow G \\ BC \rightarrow D \\ CG \rightarrow B \\ CG \rightarrow D \\ AC \rightarrow B \\ CE \rightarrow A \\ CE \rightarrow G \end{Bmatrix}$$

Step 2: Eliminating extraneous attributes on LHS. Next, remove any extraneous attributes from the left-hand side of each dependency.

Eliminating extraneous attributes:

AC → G: No extraneous attributes

Check A $(C)^+ = \{C\}$ A Is not extraneous	Check C $(A)^+ = \{A\}$ C Is not extraneous

BC → D: No extraneous attributes

Check B $(C)^+ = \{C\}$ B Is not extraneous	Check C $(B)^+ = \{B\}$ C Is not extraneous

CG → BD: No extraneous attributes

Check C $(G)^+ = \{G\}$ C Is not extraneous	Check G $(C)^+ = \{C\}$ G Is not extraneous

$AC \rightarrow B$: No extraneous attributes

Check A $(C)^+ = \{C\}$ =A Is not extraneous	Check C $(A)^+ = \{A\}$ C Is not extraneous

$CE \to A$ No extraneous attributes

Check C $(E)^+ = \{E\}$ C Is not extraneous	Check E $(C)^+ = \{C\}$ E Is not extraneous

$CE \to G$ No extraneous attributes

Check C $(E)^+ = \{E\}$ C Is not extraneous	Check E $(C)^+ = \{C\}$ E Is not extraneous

$$FDs = \left\{ \begin{matrix} AC \to G \\ D \to E \\ D \to G \\ BC \to D \\ CG \to B \\ CG \to D \\ AC \to B \\ CE \to A \\ CE \to G \end{matrix} \right\}$$

Step 3. Eliminate redundant functional dependencies

First, we eliminate any redundant dependencies by checking if any dependency can be derived from others.

Checking for redundancy:

$AC \to G$ $(AC)^+ = \{AC\}$ $= \{ACBDEG\}$ Since G appeared $AC \to G$ Redundant	$D \to E$ $(D)^+ = \{D\}$ $= \{DG\}$ $D \to E$ No redundant FD.	$D \to G$ $(D)^+ = \{D\}$ $= \{DE\}$ $D \to G$ No redundant FD.
$BC \to D$ $(BC)^+ = \{BC\}$ $BC \to D$ No redundant FD.	$CG \to B$ $(CG)^+ = CG$ $= \{CGDEAB\}$ Since B appeared $CG \to B$ Redundant	$CG \to D$ $(CG)^+ = \{CG\}$ $CG \to D$ No redundant FD.
$AC \to B$ $(AC)^+ = \{AC\}$ $AC \to B$ No redundant FD.	$CE \to A$ $(CE)^+ = \{CE\}$ $= \{CEG\}$ $= \{CEGD\}$ $CE \to A$ No redundant FD.	$CE \to G$ $(CE)^+ = \{CE\}$ $= \{CEA\}$ $= \{CEAB\}$ $= \{CEABD\}$ $= \{CEABDG\}$ Since G appeared $CE \to G$ Redundant

$$Fc = \left\{ \begin{matrix} D \to E \\ D \to G \\ BC \to D \\ CG \to D \\ AC \to B \\ CE \to A \end{matrix} \right\}$$

Step 4: Apply Union. Combine dependencies and finalize the canonical cover. ow, combine the simplified dependencies:

$$\begin{matrix} D \to E \\ D \to G \end{matrix} = D \to EG$$

Final Canonical (Minimal) Cover: The canonical (minimal) cover of F is:

$$Fc = \left\{ \begin{matrix} AC \to B \\ D \to EG \\ BC \to D \\ CG \to D \\ CE \to A \end{matrix} \right\}$$